



Anna Cai (Attempt 1 of 1)

Test Information

QUESTION 1: CALCULATED NUMERIC

5

out of 5 points

Determine the temperature reading corresponding to a 10mA analog signal from a temperature transmitter that has a span of 200 °C and a Low Range value equal to 20 °C



Given Answer: 95

Correct Answer: 95

Answer range +/- 0 (95 - 95)

QUESTION 2: CALCULATED NUMERIC

5

out of 5 points

A controller sends an 8mA analog signal to an I/P converter. The pressure signal from the I/P converter is sent to the control valve. What is the valve opening as a percentage of the total travel, assuming total travel corresponds to 100% (fully open valve)



Given Answer: 25

Correct Answer: 25

Answer range +/- 0 (25 - 25)

QUESTION 3: MULTIPLE CHOICE

5

out of 5 points

A device that converts energy into motion is called



Given Answer: Actuator

Correct Answer: Actuator

Answer:

QUESTION 4: MULTIPLE CHOICE

5

out of 5 points

A device that responds to a stimulus such as heat or pressure and generates a signal that can be measured or interpreted is called



Given Answer: Sensor

Correct Answer: Sensor

Answer:

QUESTION 5: MULTIPLE CHOICE

5 out of 5 points

A device that converts a sensor reading to a standard electrical signal is called

- Given

☒ Transmitter
- Answer:
- Correct

☒ Transmitter
- Answer:

QUESTION 6: MULTIPLE CHOICE

5 out of 5 points

The temperature in a gas fired furnace ranges between 1400 and 1500 °C. What kind of sensor device will you use to measure it

- Given

☒ Thermocouple
- Answer:
- Correct

☒ Thermocouple
- Answer:

QUESTION 7: CALCULATED NUMERIC

5 out of 5 points

What is the value of the geometry factor F_p for a **4in** fully open control valve in a **NPS 6in Schedule 40**, liquid pipeline. Use the data in **Table 1** in the CHE Calculator. **Round your numerical answer into three decimal digits.**

- Given Answer:

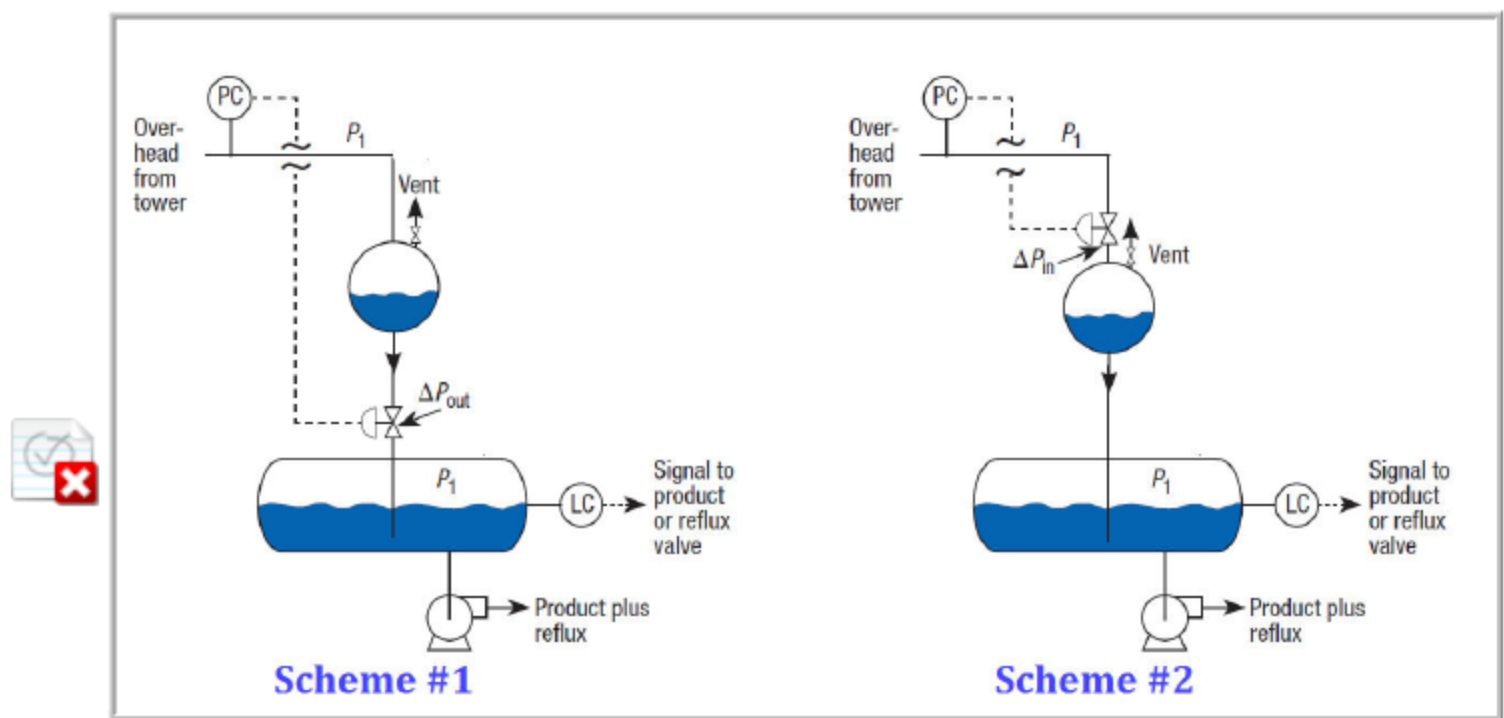
☒ .964
- Correct Answer:

☒ 0.964
- Answer range +/- 0.001 (0.963 - 0.965)

QUESTION 8: MULTIPLE CHOICE

0 out of 5 points

Which pressure control scheme from the the ones shown below would be the preferable



- Given

☒ Scheme #2
- Answer:

Answer:

Correct  Scheme #1

Answer:

QUESTION 9: CALCULATED NUMERIC

15

out of 15 points

Determine the Open Stem Travel in % of a 4in control valve with inherent characteristic given in **Table 1** in the **CHE Calculator** that is used to regulate the flow of liquid **n-butanol**. The pipeline diameter is 8in Schedule 40. The service conditions are: Flowing temperature $T=50^{\circ}\text{C}$, volumetric flow rate $Q=100\text{ m}^3/\text{hr}$, upstream pressure $P_1=1.5\text{ bar}$ and downstream pressure $P_2=1.1\text{ bar}$. Express your result in **percent travel** (for instance if your answer is 25.3% enter the value 25.3). Notice that Equations of State give very poor predictions of the thermal properties of alcohols. Round your numerical answer **into a single decimal digit**.



Given Answer:  89.8

Correct Answer:  89.9

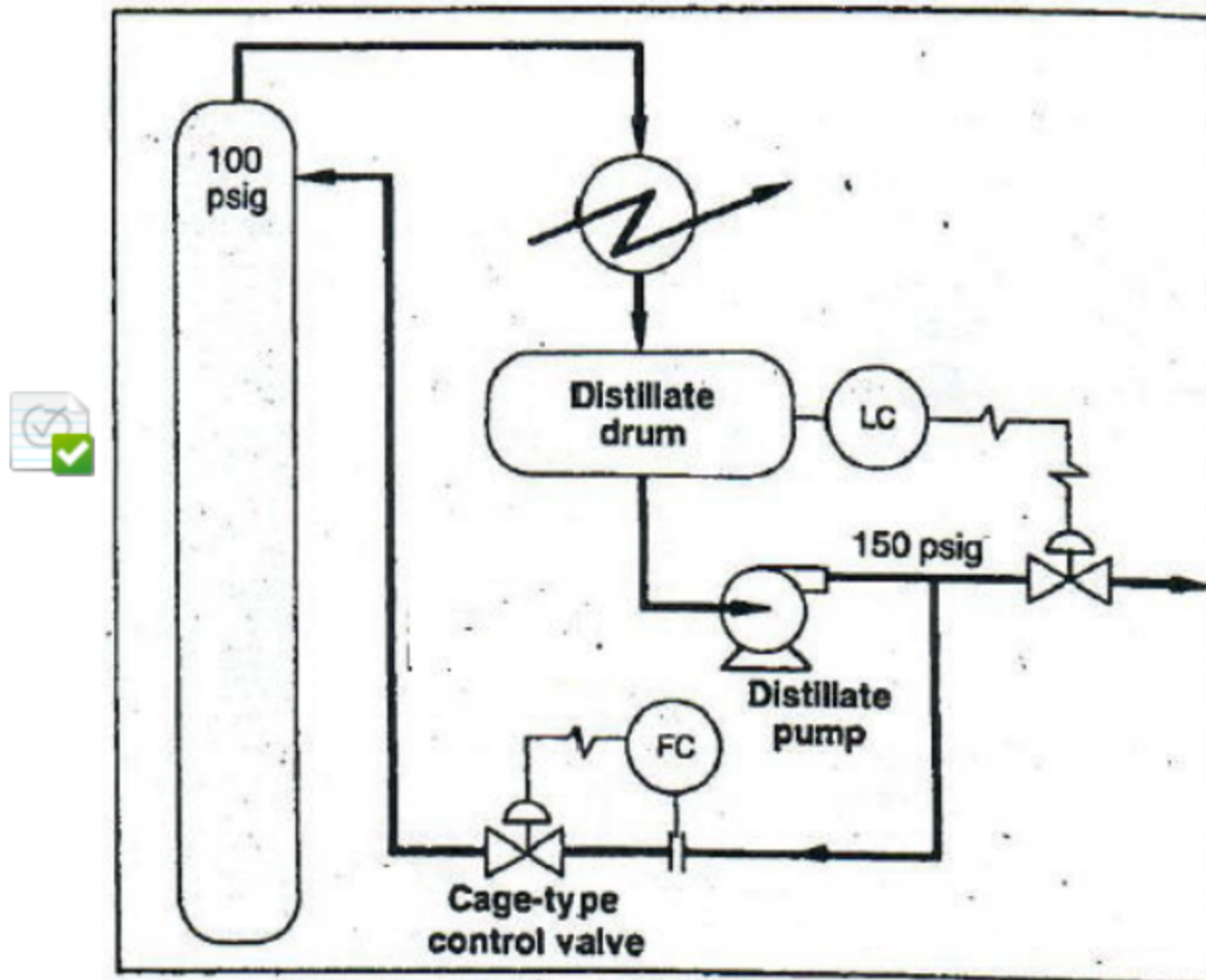
Answer range +/- 0.1 (89.8 - 90.0)

QUESTION 10: CALCULATED NUMERIC

15

out of 15 points

The system in which the reflux control valve is located starts at the pump discharge where $P_1=150$ psig. The end of the system is the fractionation column (see diagram below) that is operating at $P_2=100$ psig. The static head pressure is 19 psi and the friction drop (excluding the control valve) is 18 psi. What is the valve authority in this system. Express your numerical result as **fraction** and round your numerical answer **into two decimal digits**.



Given Answer: ✓ 0.42

Correct Answer: ✓ 0.42

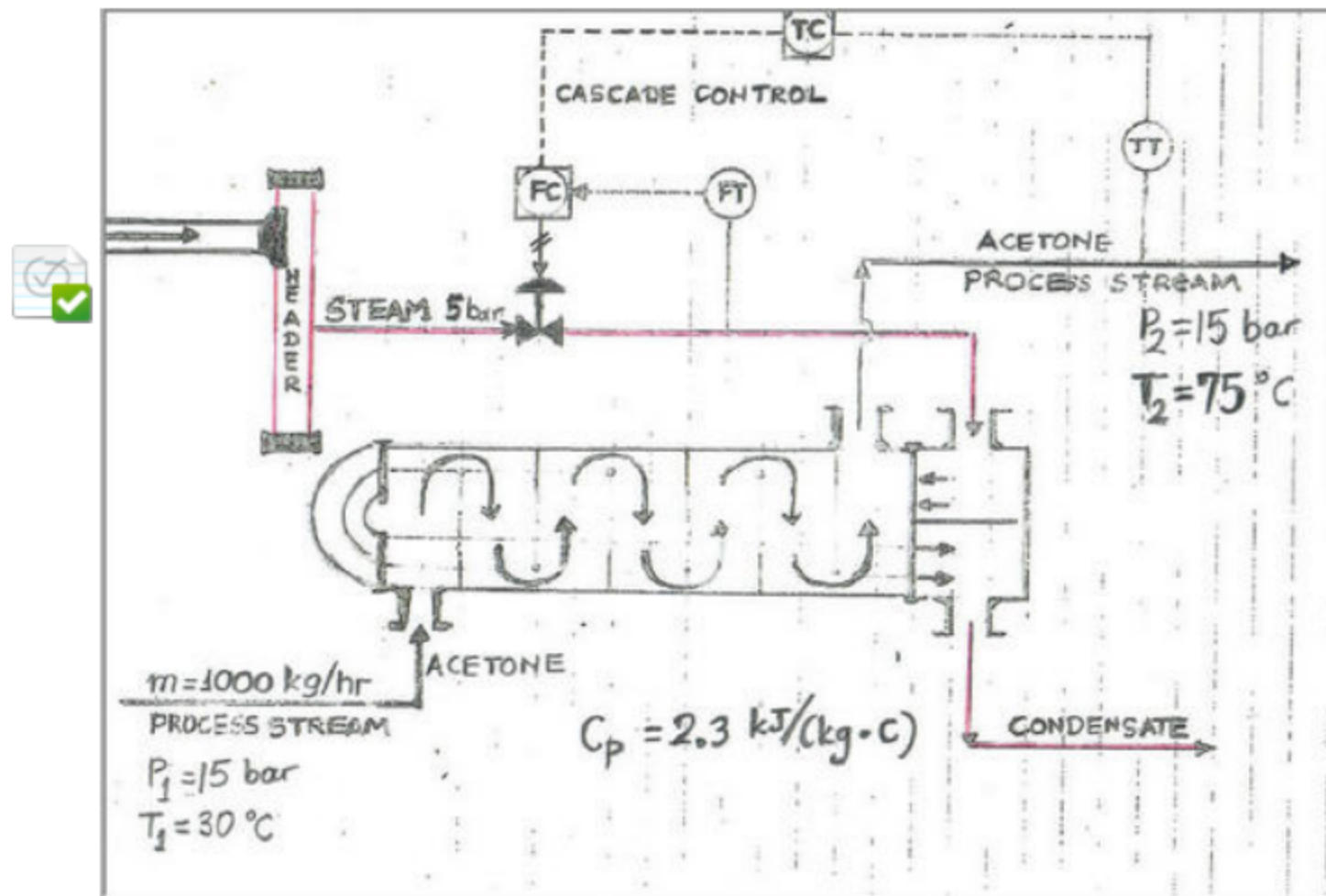
Answer range +/- 0.01 (0.41 - 0.43)

QUESTION 11: CALCULATED NUMERIC

15

out of 15 points

Cascade control is used to control the temperature of acetone in a heat exchanger using 5 bar saturated steam with quality 0.95[See sketch below]. The flow rate of acetone is 1000 kg/hr and its heat capacity is constant at 2.3 kJ/(kg °C). The target temperature is 75 °C and acetone enters the heat exchanger at 30 °C. If the flow rate of acetone drops by 10% due to an upstream pressure disturbance, what is the value of the signal in **mA** that the temperature controller is going to send in the Flow controller. Assume that the Low Range Value (LRV) of the steam flow rate is 20 kg/hr and the span 60 kg/hr. You may assume that the pressure drop in the steam valve remains constant at 3.5 bar and that the pressure drop of steam and acetone inside the heat exchanger is zero. Round your answer **into a single decimal digit**.



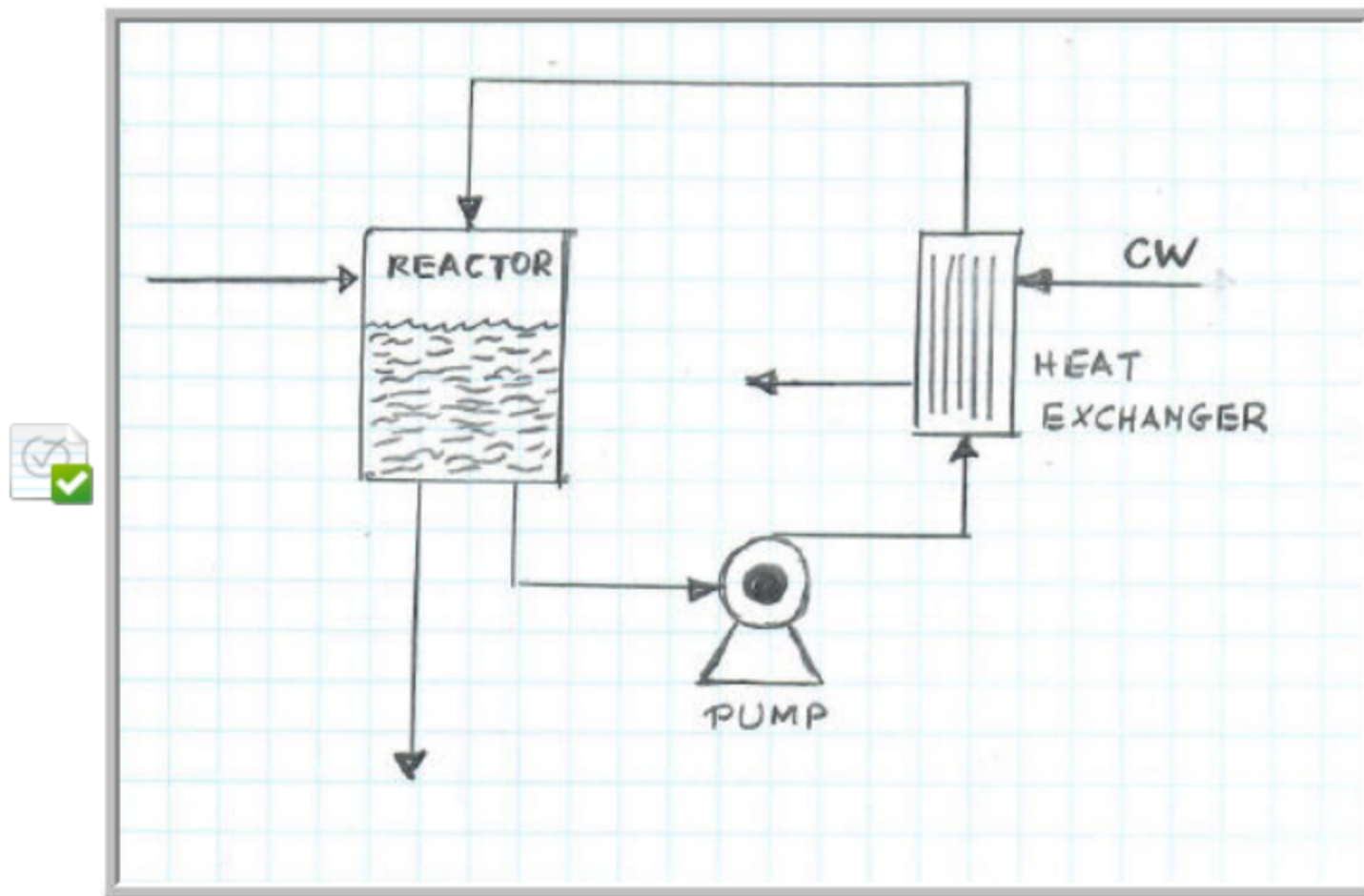
Given Answer: 10.1

Correct Answer: 10.1

Answer range +/- 0.1 (10.0 - 10.2)

QUESTION 12: CALCULATED NUMERIC**15****out of 15 points**

A chemical reaction takes place in liquid phase in a CSTR. The reaction is exothermic and the reactor temperature is regulated by circulating a portion of the product through an external heat exchanger where it is being cooled using cooling water. A schematic of this process is shown below. Calculate the Control Degrees of Freedom. Assume that the circulation pump is single speed.



Given Answer: ✓ 4

Correct Answer: ✓ 4

Answer range +/- 0 (4 - 4)